Attachment B

February 24, 2023 Ecora File No.: 230054

The Town of Ladysmith
447 Columbia St.
Kamloops, BC V2C 2T3

Attention: **Ryan Bouma, P.Eng.**

Reference: **Stocking Lake Dam – Short Term Remedial Measures**

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1. Introduction

Ecora Engineering & Resource Group Ltd. (Ecora) was retained by the Town of Ladysmith (ToL) to provide emergency engineering services with respect to a Level 1 Emergency declared on January 9, 2023 for the Stocking Lake Dam (D720127-00) located near Saltair, BC.

The Level 1 Emergency was declared due to increased seepage discharging observed exiting the downstream face of the dam at the right abutment and softening of the dam embankment material at this location during scheduled dam surveillance conducted by the ToL public works staff. The increased seepage was noted following a period of prolonged precipitation and rapid snow melt during the last week of December 2022.

Michael J. Laws, P.Eng., a Senior Dam Safety Engineer with Ecora undertook an emergency site reconnaissance on January 10, 2023, in conjunction with Ministry of Forests (MoF) Senior Regional Dam Safety Officer David Johnson, MoF Senior Dam Safety Project Manager Jeff Grass, P.Eng. and representatives from the ToL and the Regional District of Cowichan Valley.

During the emergency site reconnaissance, it was also noted that a tension crack had formed along the downstream crest of the dam.

A summary of the dam surveillance observations and climatic conditions leading up to the declaration of the Level 1 Emergency and recommendations for enhanced monitoring and emergency mitigation measures was provide to the ToL in a letter report prepared by Ecora dated February 2, 2023.

It is understood that due to poor historical dam performance it is proposed to replace Stocking Lake Dam with a new structure downstream, however this is anticipated to be at least 5 years away.

The scope of this memo is to provide recommendations for short term remedial measures to address the seepage discharging the downstream face of the dam and the area of instability until such time that the dam is replaced.

2. Proposed Short Term Remedial Measures

As the area of marginal stability along the downstream slope of the dam where the tension crack was observed has been postulated to have occurred due to a combination of rapid snow melt, precipitation and surface storm water runoff originating from the left abutment, it is proposed to construct a rock blanket, incorporating geotextile and a seepage collection pipe, along the downstream face of the dam. The purpose of the rock blanket is to mitigate the seepage exiting the downstream face of the dam at the right abutment and to provide additional stability to the downstream face, until such time as the dam is replaced.

As it is likely that the seepage discharging form the downstream face of the dam at the right abutment is coming from the spillway channel it is also proposed to remove the existing riprap from the channel and install a liner in the

base of the channel. The linear is to comprise a 6 mil PVC membrane placed between two layers of non-woven geotechnical and pinned to the top of the channel banks. Once the linear is installed the riprap would be placed back into the channel.

Details of the proposed short term remedial measures is shown on the attached drawings.

3. Rock Blanket Stability Analysis

3.1 Acceptance

The CDA Technical Bulletin, Geotechnical Consideration for Dam Safety, provides accepted minimum slope stability factors of safety for various static and seismic loading conditions for embankments as reproduced in Table 3.11.a and Table 3.1.b.

Table 3.11.a Acceptable Factors of Safety for Embankment Stability – Static Assessment

Loading Conditions	Minimum Factory of Safety	Slope
End of construction before reservoir filling	1.3	Upstream and downstream
Long-term (steady state seepage, normal reservoir level)	1.5	Downstream
Full or partial rapid drawdown	1.2 to 1.3	Upstream

Table 3.1.b Acceptable Factors of Safety for Embankment Stability – Seismic Assessment

Loading Conditions	Minimum Factory of Safety	Slope
Pseudo-Static	1.0	Upstream and downstream
Post-Earthquake	1.2 to 1.3	Upstream and downstream

3.2 Methodology

Limit Equilibrium (LE) stability analyses were carried out to evaluate the static stability of the downstream face of the dam under existing conditions and incorporating the proposed rock blanket.

Circular (global) factors of safety for the proposed slopes were calculated using the two-dimensional LE analysis program Slide2.0 (version 9.022 by Rocscience) utilizing the Morgenstern-Price method with a half sine interslice force adopted.

The cross-sectional geometry used for the analysis was taken from previous topographical survey data obtained for the conceptual design of the replacement dam (Ecora, 2021), where a critical section in the area of marginal stability was analyzed. In the slope model, the stratigraphy and geotechnical parameters utilized in the analysis was derived from historical boreholes drilled through the embankment.

A coupled two-dimensional steady-state FEA seepage and slope stability back analysis (FoS ~1.0), modelling increased pore water pressure conditions in the downstream face of the embankment during the period of increased snow melt, precipitation, and stormwater runoff, was undertaken to determine pore water conditions to be used in the design of the rock blanket. The increased precipitation and stormwater runoff was modelled with a vertical infiltration boundary condition along the crest and downstream face of the dam.

3.3 Soil Parameters

Geotechnical design parameters used in the analysis are given in Table 3.3 below and have been derived based on the results of the site investigation of the embankment materials previously undertaken by Ecora (2018 & 2021), published data, and Ecora's previous local experience.

The analysis generally assumes that the soil shear strength is governed by the Mohr-Coulomb criteria given by:

$$\tau = c' + \sigma' tan\phi'$$

Where: τ = Soil Shear Strength;
c' = Effective Cohesion;
σ' = Normal Effective Stress; and,
ϕ' = Effective Angle of Internal Friction.

The riprap rockfill material has been modelled using the Snowden Modified Anisotropic non-linear strength model as discussed in Section 3.4 below.

Table 3.3 Summary of Input Parameters

Material Name	Strength Type	Unit Weight (kN/m^3)	Cohesion (kPa)	Friction Angle (°)
Embankment Fill	Mohr-Coulomb	19	0.5	32
Riprap	Snowden Modified Anisotropic Linear	24	-	-

3.4 Rock Fill Strength

Ecora carried out stability analysis using a non-linear strength model for rock fill materials (Douglas, 2003) where the rock fill strength is dependent on the normal stress (σ_n), fines content, angularity rating, and intact rock Unconfined Compressive Strength (UCS) of the rockfill.

Douglas (2003) established a large database of published and unpublished tri-axial testing undertaken on rock fill and carried out an extensive statistical analysis of the factors affecting its strength.

The database determined that a non-linear relationship exists between the secant friction angle (Φ'_{sec}) of the rock fill and normal stress (σ_n). Based on a non-linear statistical analysis of the database it was determined that the angularity rating, percentage of fines (FINES), coefficient of curvature (cc) and the UCS of the rock fill have the most influence on the Φ'_{sec} with the following equation developed to express the relationship between Φ'_{sec} and σ_n.

$$\Phi' sec = a + b\,\sigma'^{c}_{n}$$

Where:

a = 36.43 - 0.267ANG - 0.172FINES + 0.756(c_c - 2) + 0.0459(UCS - 150)

b = 69.51 + 10.27ANG + 0.549FINES - 5.10(c_c - 2) - 0.408(UCS - 150) - 0.408

ANG = (angularity rating - 5) for angularity rating >5.5, otherwise ANG = 0

Based on lower bound parameters for andesitic rockfill comprising an angularity rating of 5, coefficient of curvature of 2, fines content of 5%, and an intact UCS of 35 MPa, a non-linear envelope was developed for the analysis.

3.5 Results

The results of the stability analysis are summarized in Table 3. below. Static slope stability results are presented in Figures 1 and 2.

Ecora recommends that additional slope stability analyses be conducted if any significant change in embankment geometry occurs or if a change in conditions is observed during construction.

Table 3.5 Global Stability Results

Model Scenario	Calculated Static Minimum FoS	Figure
Existing Condition Back Analysis	1.10	1
Rock	1.53	2

4. Geotextile Filter Design

As discussed in Fell et. Al. (2015) geotextiles can be used as filters in smaller dam's provided that they;

- Prevent erosion of soil particles from the base soil they are protecting, i.e. act as a filter,
- Allow drainage of seepage water without significant flow impedance;
- Not clog excessively with particles from the soil they are protecting;
- Be sufficiently strong to resist damage during construction, and;
- Be sufficiently durable to resist adverse ultraviolet light and chemical exposure during construction in service.

Fell et. al. (2015) further discusses that there are many different filter design criteria utilized for geotextiles, however there are two methods broadly used in North America, namely the method described in Luettich et. al. (1992) as recommended by FEMA (2008) and the Federal Highway Administration (FHWA) method (2008).

Ecora utilized the Luettich et. al. (1992) method for determining a suitable geotextile in the construction of the rock fill blanket.

The Luettich et. al. (1992) method provides two design criteria one for steady state (standard drainage and collection systems) flow conditions and the other for dynamic (shoreline and coastal application) flow conditions as decision tree diagrams.

Using the decision tree diagram for steady state flow conditions (Figure 3) and the results of previous laboratory testing undertaken on the embankment fill materials (Ecora 2018 and 2021) the geotextile should have an apparent opening size (O_{95}) of less than 0.226 mm.

5. Construction

5.1 General

The proposed remedial measures shall not be constructed during the spring freshet or any high-volume precipitation event. Prior to commencement of construction of the remedial measures, the lake shall be drawn down approximately 1.0 m from full pool to allow for saturated areas to dewater as practically as possible and to provide a safety buffer in the event that a non-predicted rainfall event occurs during construction. The

Environment Canada weather forecast shall also be consulted prior to commencement of construction to ensure that there are no major periods of rainfall anticipated during construction of the short-term remedial measures.

A typical cross-section and plan view depicting the proposed short-term remedial measures are attached in Appendix B.

5.2 Rock Fill Blanket

The following general considerations shall be undertaken during construction of the proposed rock fill blanket:

- All vegetation and topsoil shall be removed from the footprint of the proposed rock fill blanket;
- All removed vegetation and topsoil shall be disposed away from the dam;
- Any minor hollows should be infilled with a well graded sand and well tamped-in prior to placement of the geotextile;
- The geotextile shall be placed lengthways down the face of the dam (roll direction) with a minimum overlap of 300 mm between adjacent sections.
- The geotextile shall be temporarily pinned to the face of the dam prior to the placement of rock fill; and,
- Placement of rock fill shall not be end-dumped. The rock shall be machine manipulated for satisfactory setting and spacing and placed from the dam toe to crest direction.

5.3 Spillway Channel Liner

The following general considerations shall be undertaken during construction of the proposed spillway channel liner:

- The existing riprap shall be removed from the section of the channel to be lined;
- The base of the channel shall be inspected for pipes, hollows, and voids;
- If any pipe or voids are encountered along the embankment side of the channel these shall be back filled with dry bentonite chips prior to placement of the lower section of geotextile;
- Any minor hollows should be infilled with a well graded sand and well tamped-in prior to placement of the lower section geotextile;
- The geotextile shall be placed lengthways down the spillway channel (roll direction) with a minimum overlap of 500 mm between adjacent sections.
- The lower sections of geotextile shall be temporarily pinned to the spillway channel prior to the placement of PVC liner;
- The PVC liner shall be of sufficient dimension that no joins are required and the top of the liner shall be pinned to the crest of the spillway channel prior to the placement of the upper geotextile;
- The upper section of geotextile shall be pinned to the crest of the spillway channel prior to the placement of the riprap; and,
- The riprap shall not be end-dumped back into the spillway channel and shall be machine manipulated for satisfactory setting and spacing. It should be placed along the channel base first

from an upstream to downstream direction and then along the sides again in an upstream to downstream direction.

5.4 Recommended Materials

The following materials recommended for use in the construction of the remediation works shall consist of the following,

- Free draining rock fill material such as Class 25 to 50 kg riprap that generally conforms to the requirements of Table 205A and 205C of the BC MoTI Standard Specifications for Highway Construction (2020);
- A non-woven geotextile fabric placed on the prepared subgrade soils, underneath the rock blanket and riprap materials with an apparent opening size (O95) of less than 0.226 mm. Based on Ecora's experience with similar armouring designs, a Nilex 4553 (or approved equivalent on the BC MoTI recognized product list) is an acceptable non-woven geotextile filter fabric to be placed at the soil-riprap or rock blanket interface; and'
- A PVC liner (sheet) with a minimum thickness of 6 mil that meets the ASTM-D-5035 and ASTM-D-02261 standards with regards to tear resistance.

Both the riprap and geotextile shall be placed in accordance with the drawings in Appendix B.

6. Limitations of Report

This report and its contents are intended for the sole use of the Town of Ladysmith, their agents and the applicable regulatory authorities. Ecora Engineering & Resource Group Ltd. (Ecora) does not accept any responsibility for the accuracy of any data, analyses, or recommendations contained or referenced in the report when the report is used or relied upon by any Party other than the Town of Ladysmith, their agents, the applicable regulatory authorities or for any Project other than that described in this report. Any such unauthorized use of this report is at the sole risk of the user.

Where Ecora submits both electronic file and hard copy versions of reports, drawings, and other project-related documents, only the signed and/or sealed versions shall be considered final and legally binding. The original signed and/or sealed version archived by Ecora shall be deemed to be the original for the Project. Both electronic file and hard copy versions of Ecora's deliverables shall not, under any circumstances, no matter who owns or uses them, be altered by any party except Ecora.

Ecora's General Conditions are provided in Appendix A of this report.

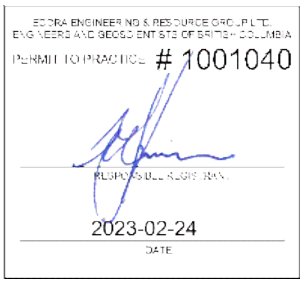
7. Closure

We trust this report meets your requirements. If you have any questions or comments, please contact the undersigned.

Sincerely
Ecora Engineering & Resource Group Ltd.

Prepared & Approved by:





Reviewed by:



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Attachments: Tables
Figures
Photos
Appendix A General Conditions
Appendix B Drawings

Additional Optional Information:

Version Control and Revision History

Version	Date	Prepared By	Reviewed By	Approved By	Notes/Revisions
0	2023-02-04	MJL	AJK	MJL	Issued for Use

References

BC Ministry of Transportation and Infrastructure, 2020. Standard Specification for Highway Construction.

BC Ministry of Transportation and Infrastructure, January 2023. Recognized Products List.

Canadian Dam Association (CDA), 2007. Dam Safety Guidelines 2013 Edition.

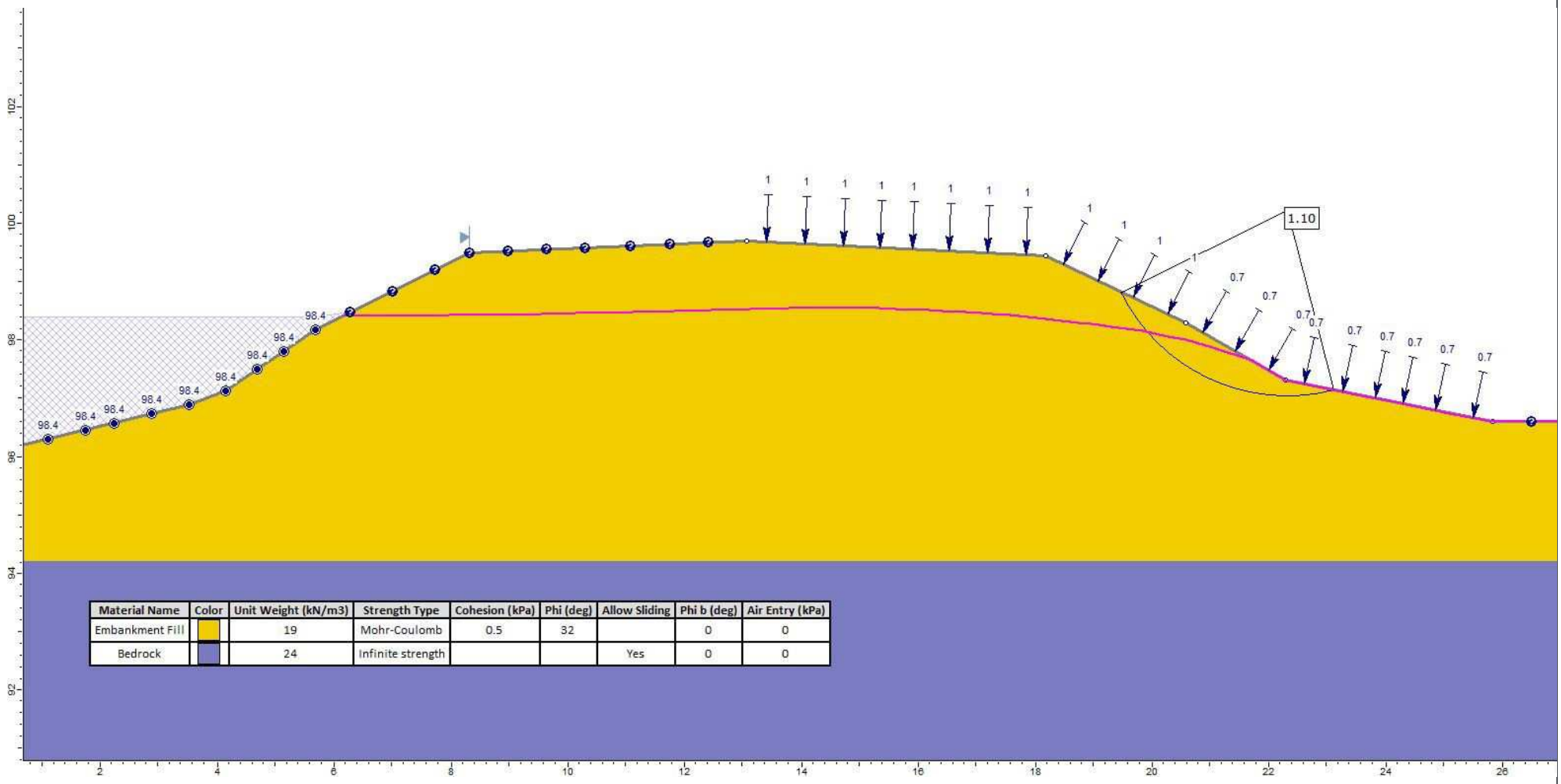
Ecora Engineering & Resource Group Ltd., November 2018. Dam Safety Review and Risk Assessment of Stocking Lake Dam.

Ecora Engineering & Resource Group Ltd., August 2021. Stocking Lake Dam Remediation Conceptual Design.

Fell, R., MacGregor, P., Stapledon, D., Bell, G., and Foster, M.., 2018. Geotechnical Engineering of Dams, 2nd Edition.

Figures

Figure 1` Downstream Slope Stability Analysis – Existing Conditions
Figure 2` Downstream Slope Stability Analysis – Rock Blanket Addition
Figure 3` Geotextile Design Selection Based on Steady-State Flow Conditions



Material Name	Color	Unit Weight (kN/m3)	Strength Type	Cohesion (kPa)	Phi (deg)	Allow Sliding	Phi b (deg)	Air Entry (kPa)
Embankment Fill		19	Mohr-Coulomb	0.5	32		0	0
Bedrock		24	Infinite strength			Yes	0	0

Notes:

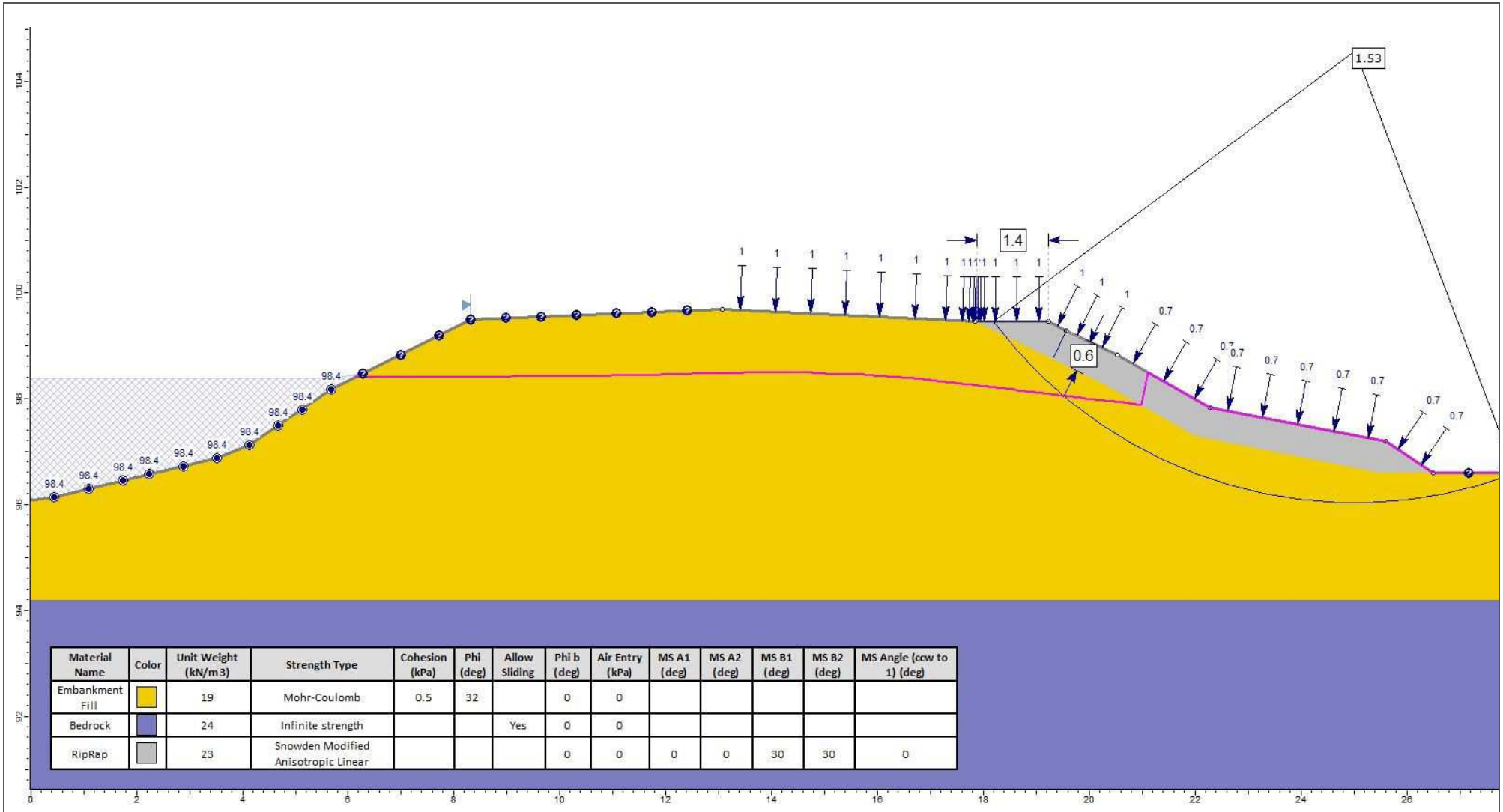
STOCKING LAKE DAM EMERGENCY RESPONSE

Downstream Slope Stability Analysis – Existing Conditions

Project No. 230054
Client: Town of Ladysmith
Office: Kelowna
Scale: NTS
Date: Feb 09, 2023
DWN: DA CHK: MJL

ecora

Figure 1.0



Material Name	Color	Unit Weight (kN/m3)	Strength Type	Cohesion (kPa)	Phi (deg)	Allow Sliding	Phi b (deg)	Air Entry (kPa)	MS A1 (deg)	MS A2 (deg)	MS B1 (deg)	MS B2 (deg)	MS Angle (ccw to 1) (deg)
Embankment Fill		19	Mohr-Coulomb	0.5	32		0	0					
Bedrock		24	Infinite strength			Yes	0	0					
RipRap		23	Snowden Modified Anisotropic Linear				0	0	0	0	30	30	0

Notes:

STOCKING LAKE DAM EMERGENCY RESPONSE

Downstream Slope Stability Analysis – Rock Blanket Addition

Project No. 230054
Client: Town of Ladysmith
Office: Kelowna
Scale: NTS
Date: Feb 09, 2023
DWN: DA CHK: MJL

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Figure 2.0

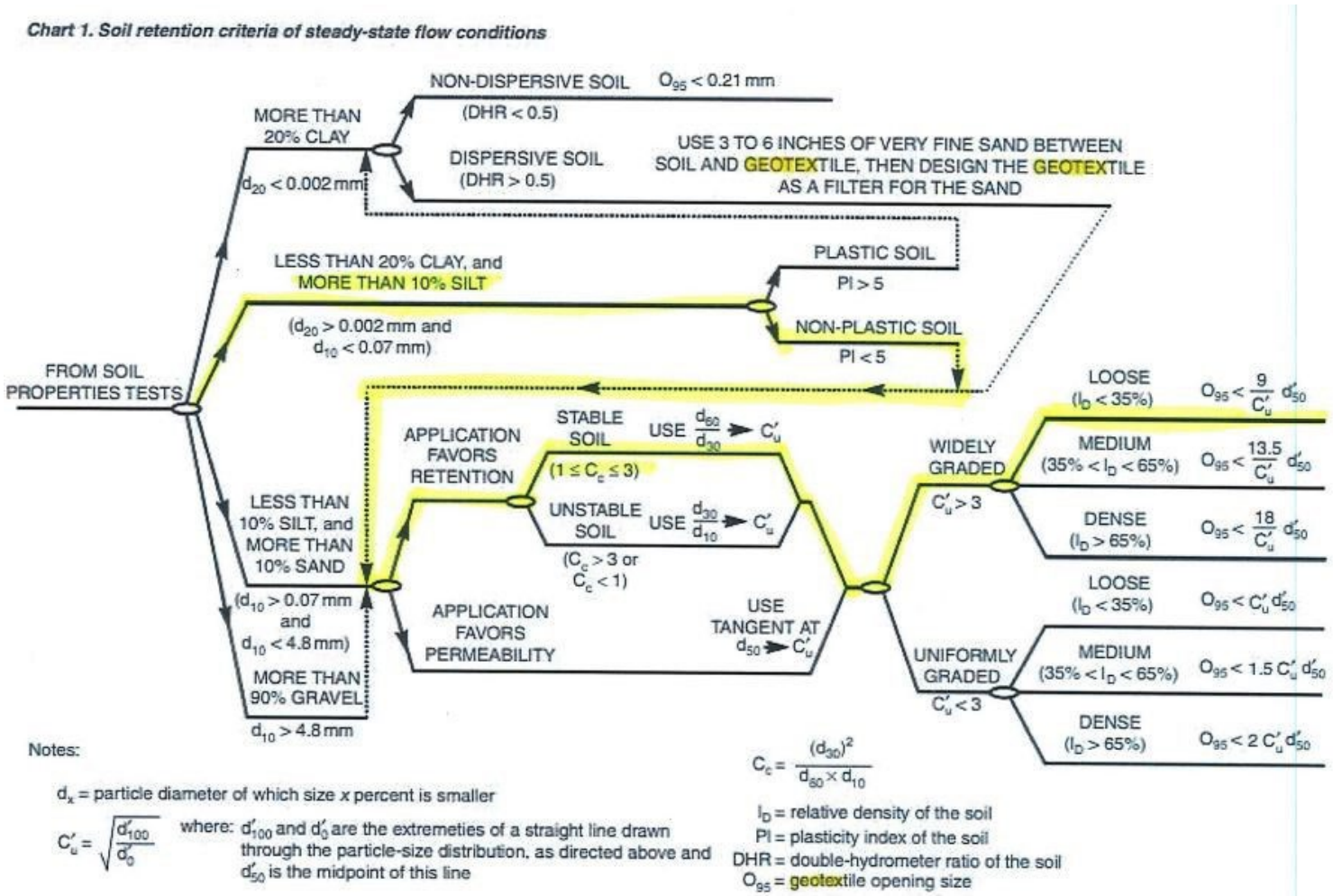


Figure 9.33(a) **Soil retention criteria for steady-state flow conditions (Luettich *et al.*, 1992).**

Notes:
Soil Retention Criteria chart referenced in Geotechnical Engineering of Dams, 2nd Edition, 2015.

STOCKING LAKE DAM EMERGENCY RESPONSE

Geotextile Design Selection Based on Steady-State Flow Conditions

Project No.	230054
Client:	Town of Ladysmith
Office:	Kelowna
Scale:	NTS
Date:	Feb 13, 2023
DWN:	DA CHK: MJL

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Figure 3.0

Appendix A

General Conditions

1. Standard of Care

Ecora Engineering and Resource Group Ltd. (Ecora) has prepared this report in a manner consistent with that level of care and skill ordinarily exercised by members of the engineering and science professions currently practicing under similar conditions in the jurisdiction in which the services are provided, subject to the time limits and physical constraints applicable to this report. No other warranty, expressed or implied is made.

2. Basis and Use of the Report

This report and the recommendations contained in it are intended for the sole use of Ecora's Client, their agents and the applicable regulatory authorities. Ecora does not accept any responsibility for the accuracy of any of the data, the analyses or the recommendations contained or referenced in the report when the report is used or relied upon by any party other than Ecora's Client, their agents and the applicable regulatory authorities unless otherwise authorized in writing by Ecora. Any unauthorized use of the report is at the sole risk of the user. In order to properly understand the suggestions, recommendations and opinions expressed herein, reference must be made to the whole of the report. We cannot be responsible for use by any party of portions of the report without reference to the whole report.

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3. Alternate Report Format

Where Ecora submits both electronic file and hard copy versions of reports, drawings and other project-related documents, only the signed and/or sealed versions shall be considered final and legally binding. The original signed and/or sealed version archived by Ecora shall be deemed to be the original for the Project. Both electronic file and hard copy versions of Ecora's deliverables shall not, under any circumstances, no matter who owns or uses them, be altered by any party except Ecora.

4. Soil, Rock and Groundwater Conditions

Classification and identification of soils, rocks and geological units have been based upon commonly accepted systems and methods employed in professional geotechnical practice. This report contains descriptions of the systems and methods used. Classification and identification of the type and condition of these materials or units involves judgment, and boundaries between different soil, rock or geologic types or units may be transitional rather than abrupt. Accordingly, Ecora does not warrant conditions represented herein as exact, but infers accuracy only to the extent that is common in practice.

Soil and groundwater conditions shown in the factual data and described in the report are the observed conditions at the time of their determination or measurement. Unless otherwise noted, those conditions form the basis of the recommendations in the report. Groundwater conditions may vary between and beyond reported locations and can be affected by annual, seasonal and meteorological conditions. The condition of the soil, rock and groundwater may be significantly altered by construction activities such as traffic, excavation, groundwater level lowering, pile driving, blasting on the site or on adjacent sites. Excavation may expose the soils to climatic elements such as freeze/thaw and wet /dry cycles and/or mechanical disturbance which can cause severe deterioration. Unless otherwise indicated the soil must be protected from these changes during construction.

5. Environmental and Regulatory Issues

The professional services retained for this project include only the geotechnical aspects of the subsurface conditions at the site, unless otherwise specifically stated and identified in the report. The presence or implication(s) of possible surface and/or subsurface contamination resulting from previous activities or uses of the site and/or resulting from the introduction onto the site of materials from off-site sources are outside the terms of reference for this project and have not been investigated or addressed.

6. Sample Disposal

Ecora will dispose all soil and rock samples for 30 days following issue of this report. Further storage or transfer of samples can be made at the Client's expense upon written request, otherwise samples will be discarded.

7. Construction Services

During construction, Ecora should be retained to perform sufficient and timely observations of encountered conditions to confirm and document that the subsurface conditions do not materially differ from those interpreted conditions considered in the preparation of Ecora's report and to confirm and document that construction activities do not adversely affect the suggestions, recommendations and opinions contained in Ecora's report. Adequate field review, observation and testing during construction are necessary for Ecora to be able to provide letters of assurance, in accordance with the requirements of many regulatory authorities. In cases where this recommendation is not followed, Ecora's responsibility is limited to interpreting accurately the information encountered at the borehole locations, at the time of their initial determination or measurement during the preparation of the Report.

8. Job Site Safety

Ecora is responsible only for the activities of our employees on the jobsite. The presence of Ecora's personnel on the site shall not be construed in any way to relieve the Client or any contractors on site from their responsibilities for site safety. The Client acknowledges that he, his representatives, contractors or others retain control of the site and that Ecora never occupy a position of control of the site. The Client undertakes to inform Ecora of all hazardous conditions, or other relevant conditions of which the Client is aware. The Client also recognizes that our activities may uncover previously unknown hazardous conditions or materials and that such a discovery may result in the necessity to undertake emergency procedures to protect our employees as well as the public at large and the environment in general.

9. Changed Conditions and Drainage

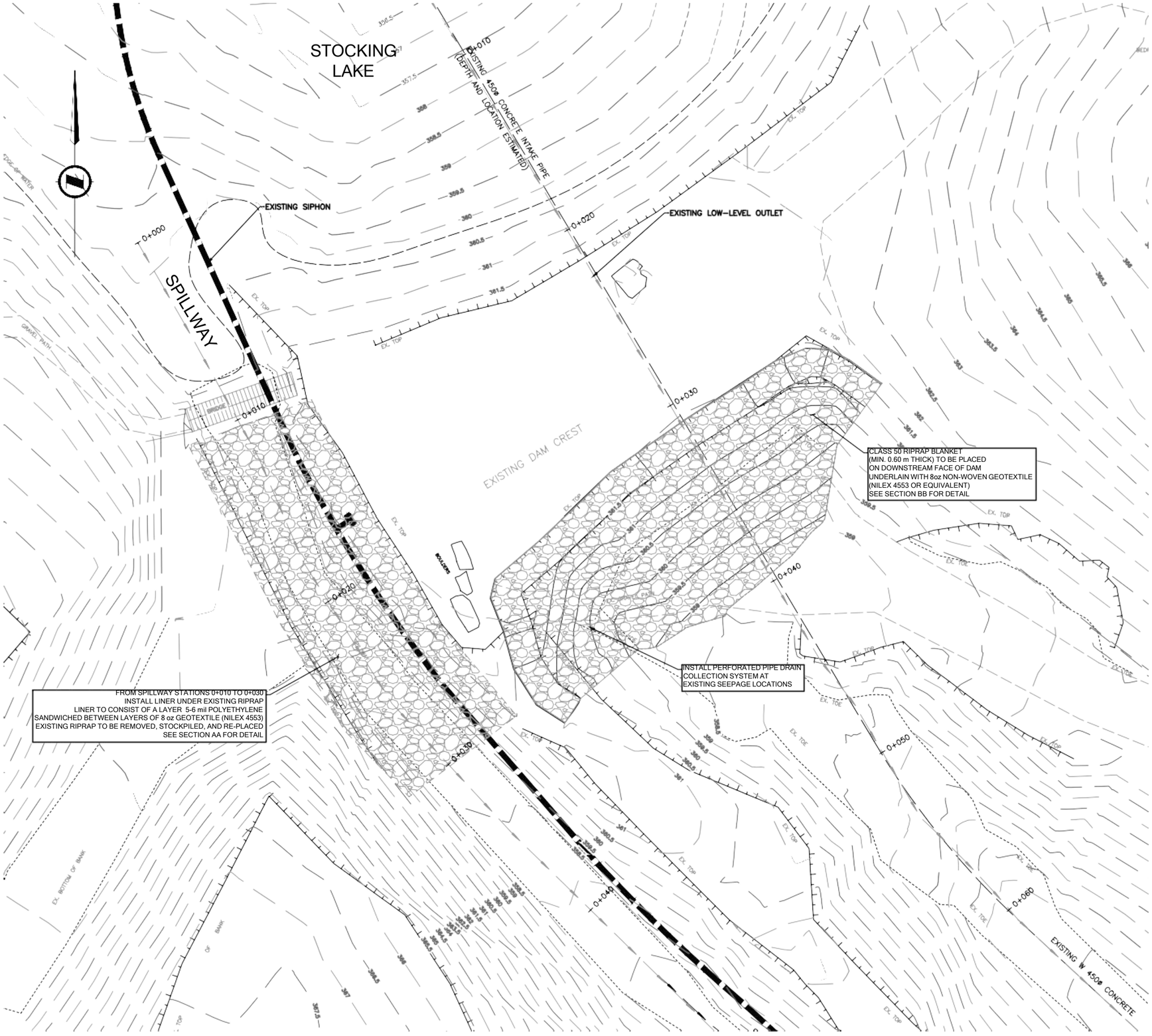
Where conditions encountered at the site differ significantly from those anticipated in this report, either due to natural variability of subsurface conditions or construction activities, it is a condition of this report that Ecora be notified of any changes and be provided with an opportunity to review or revise the recommendations within this report. Recognition of changed soil and rock conditions requires experience and it is recommended that Ecora be employed to visit the site with sufficient frequency to detect if conditions have changed significantly. Drainage of subsurface water is commonly required either for temporary or permanent installations for the project. Improper design or construction of drainage or dewatering can have serious consequences. Ecora takes no responsibility for the effects of drainage unless specifically involved in the detailed design and construction monitoring of the system.

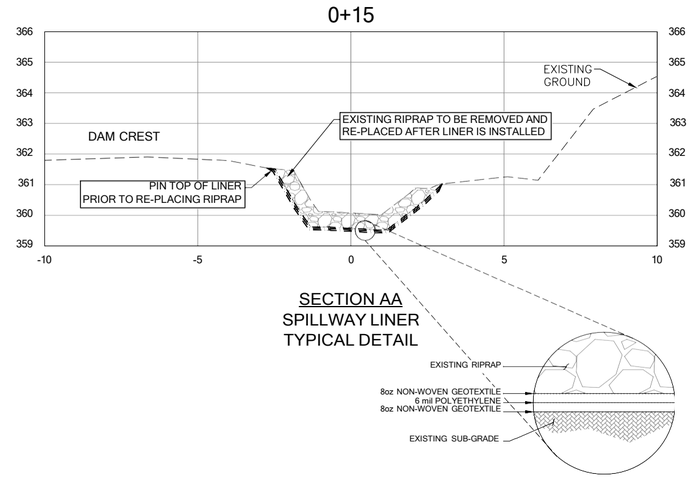
10. Services of Sub consultants and Contractors

The conduct of engineering and environmental studies frequently requires hiring the services of individuals and companies with special expertise and/or services which we do not provide. Ecora may arrange the hiring of these services as a convenience to our Clients. As these services are for the Client's benefit, the Client agrees to hold the Company harmless and to indemnify and defend Ecora from and against all claims arising through such hiring's to the extent that the Client would incur had he hired those services directly. This includes responsibility for payment for services rendered and pursuit of damages for errors, omissions or negligence by those parties in carrying out their work. In particular, these conditions apply to the use of drilling, excavation and laboratory testing services.

Appendix B

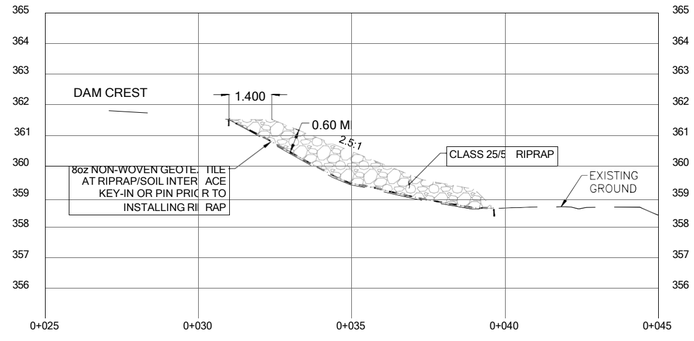
Short Term Remedial Works Drawings





SECTION AA
SPILLWAY LINER
TYPICAL DETAIL

0 1 1:100 5m



SECTION BB
ROCK BLANKET TYPICAL DETAIL

LEGEND
STREAM
BoB
ToB
EDGE OF PATH

GEOGRAPHIC PROJECTION AND DATUM:
UTM NAD83 ZONE 10N

VERTICAL DATUM:
HT v2.0 (2010) - REFERENCE CGVD28

NO.	DATE (YY/MM/DD)	DRN	REVISION	CH'KD
B	20230224	AJK	ISSUED FOR REVIEW	MJL
A	20230210	AJK	ISSUED FOR REVIEW	MJL

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SEAL

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DESIGN: MJL
DRAWN: AJK
CHECKED: MJL
DATE: 20230210
SCALE: 1:100

TOWN OF LADYSMITH

STOCKING LAKE DAM EMERGENCY REPAIRS
SITE PLAN & TYPICAL DETAILS

Original Drawing Size
ISO full bleed A1 (594.00 x 841.00 MM)

Drawing No. 230054_G1.0

Rev.No A

FILE LOCATION: Z:\2021\230054\Layout\Working\230054_Stocking Lake Dam_Plan&Profile_20230209.dwg
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